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PPG-Based Atrial Fibrillation Detection

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Abstract

Abstract Atrial fibrillation (AF) is the most common sustained arrhythmia and a major, preventable cause of ischemic stroke. While electrocardiogram (ECG) remains the diagnostic gold standard, its intermittent use can miss asymptomatic or paroxysmal AF, motivating scalable screening via wearable photoplethysmography (PPG). Large population studies using smartwatch PPG (e.g., Apple, Huawei, Fitbit) demonstrate high positive predictive value when alerts are issued conservatively, but also reveal an important limitation: current wearable approaches prioritize specificity and often sacrifice sensitivity, largely because wrist PPG is highly susceptible to motion artifacts and confounding irregular rhythms (e.g., PACs/PVCs). This project aims to narrow the performance gap between PPG-only and ECG-based AF detection by developing a robust end-to-end pipeline that improves reliability under realistic noise conditions while maintaining clinically acceptable false-alarm rates. We propose a PPG-based AF detection system combining (i) preprocessing and segmentation of 30 s windows, (ii) explicit signal-quality assessment to gate unreliable segments, and (iii) AF classification using both an interpretable feature-based baseline and a compact deep learning model. The system is trained and evaluated on datasets with simultaneous ECG labels (ground truth) and PPG inputs, using patient-wise splits to ensure generalization. Signal quality is quantified using complementary indices (spectral energy concentration in the physiological band, peak-detection plausibility checks, and beat-shape/template consistency), enabling the model to abstain when the PPG is too corrupted. In line with instructor guidance, a key novelty is targeting on-device inference: designing a lightweight model intended to run directly on the smartwatch (rather than server-side), enabling real-time screening "at the edge." We further incorporate wearable inertial sensing context—specifically IMU/accelerometer-derived motion—to inform or validate quality gating in practical deployment, while still evaluating PPG robustness where required. Performance will be assessed against ECG labels using sensitivity, specificity, PPV, F1, and ROC-AUC at the segment level, and translated into episode-level detection behavior under alerting rules. This interim report documents the literature foundation, design choices, and implementation plan for developing an artifact-aware, device-feasible AF screening pipeline that approaches ECG-level accuracy without compromising user trust.

Contents

Chapter 1

Introduction

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia and is a major risk factor for ischemic stroke, associated with about a five-fold increase in stroke risk. Its prevalence rises sharply with age, affecting over 10% of people older than 80 years [6]. Early detection and treatment of AF can significantly reduce stroke incidence and mortality. However, AF is often asymptomatic or paroxysmal (intermittent), so many cases go undiagnosed in routine clinical practice. Traditional detection relies on electrocardiogram (ECG) recordings, such as a 12-lead ECG during a doctor's visit or short-term ambulatory monitors, which can easily miss infrequent episodes. This has motivated interest in continuous or frequent monitoring using wearable devices to capture AF episodes that would otherwise be overlooked [7].

Photoplethysmography (PPG) is an optical technique used in smartwatches and fitness trackers to measure the peripheral pulse. Because each heartbeat generates a pulse wave detectable by PPG sensors, these devices can monitor heart rhythm non-invasively and continuously in everyday settings [8]. PPG-based heart rate monitoring is already widespread in consumer wearables, and recent large-scale studies have shown that such devices can alert users to possible AF. For example, the Apple Heart Study (2019) used an Apple Watch's PPG to identify irregular pulse rhythms; among approximately 419,000 participants, 0.52% received an AF notification, and subsequent ECG patch monitoring confirmed AF in about one-third of those notified [9]. The positive predictive value (PPV) of an AF alert in that study was reported around 84% for concurrent AF on a patch ECG. Similarly, the Huawei Heart Study in China screened over 187,000 people with PPG-based algorithms; about 0.23% received "suspected AF" notifications, and of those who followed up, 87% were confirmed to have AF, corresponding to a PPV around 90% [10]. More recently, the 2020 Fitbit Heart Study achieved an even higher PPV (approximately 98%) by analyzing only high-quality PPG data segments and requiring at least 30 minutes of irregular rhythm before issuing an alert [11]. These trials demonstrate the potential of wearables to detect undiagnosed AF on a broad scale, but they also highlight a key point: to avoid false alarms, current wearable algorithms are tuned for high specificity at the expense of sensitivity (e.g., the Fitbit algorithm had approximately 68% sensitivity for AF episodes during one-week validation). In practice, PPG-based AF detection still falls short of the accuracy and reliability of standard ECG-based diagnosis.

The main challenge is that PPG is an indirect measure of cardiac rhythm and is susceptible to various noise sources. Unlike ECG, which electrically captures each depolarization of the heart and clearly shows AF's hallmarks (absent P-waves and irregular R–R intervals), PPG infers heartbeats from blood volume pulses at the periphery. PPG signals lack direct information about atrial activity and can be distorted by motion, poor sensor contact, or low perfusion [8]. In fact, free-living wrist PPG recordings often contain significant motion artifacts; studies report that 20–40% of recording time may be corrupted in typical wearable data. These artifacts can mimic or obscure arrhythmias, leading to false positives or missed detections if not handled properly. Moreover, certain non-AF arrhythmias, such as premature atrial contractions (PACs) or premature ventricular contractions (PVCs), cause irregular pulse patterns that can confuse PPG-based algorithms into falsely classifying AF. Thus, achieving high sensitivity and specificity with only PPG is an ongoing technical hurdle [12].

This project aims to bridge the performance gap between PPG-based and ECG-based AF detection. In other words, we seek to improve PPG-only detection methods so that they approach the accuracy and robustness of gold-standard ECG algorithms. Building on recent advances, we propose to develop a signal processing and machine learning pipeline that can more reliably distinguish AF from normal rhythm and other arrhythmias using just the PPG waveform. Key objectives include enhancing noise artifact handling, capturing the telltale irregularity of AF amidst various confounders, and validating the approach on real-world data. By the end of the project, we will evaluate how close our PPG-based algorithm comes to ECG-level performance in terms of sensitivity (correctly detecting AF episodes) and specificity (minimizing false alarms). Such improvements could pave the way for more trustworthy AF screening on ubiquitous wearable devices, enabling earlier diagnosis and intervention for individuals with silent or intermittent AF.

In this interim report, we first review the background and related work in detail, analyzing the state-of-the-art in PPG-based AF detection and its challenges (Section Background). We then outline our proposed methodology (Section Project Plan), including data sources, algorithm design, and a timeline for implementation. We describe how we plan to evaluate success (Section Evaluation Plan), and we discuss any ethical or societal considerations (Section Ethical Issues). This report summarizes progress to date—primarily an extensive literature review and planning—and lays out the path forward for developing and testing the proposed solution in the coming months.

Chapter 2

Background and Related Work

Clinical importance of AF detection: AF is a supraventricular tachyarrhythmia characterized by disorganized atrial electrical activity and an "irregularly irregular" ventricular response. It is associated with a substantial health burden. Besides the $\sim 5\times$ increase in stroke risk, AF can lead to heart failure, cognitive decline, and increased mortality if untreated. Many AF-related strokes are preventable with appropriate anticoagulation, so early detection of AF is crucial. However, AF can be asymptomatic; studies estimate that a significant fraction of AF cases are "silent" and only discovered after a stroke or other complication. Paroxysmal AF (episodes that start and stop on their own) is especially hard to catch with infrequent ECGs [1]. Traditional monitoring methods like 24-hour Holter monitors or event recorders have limited duration. Implantable loop recorders offer long-term ECG monitoring and achieve high detection yield, but they are invasive and expensive. This context has created strong clinical momentum for non-invasive, long-term monitoring solutions.

Wearable devices with PPG sensors present an attractive approach to continuously screen for AF in the general population. PPG sensors are found in popular smartwatches (e.g., Apple Watch, Fitbit, Samsung Galaxy Watch) and even smartphone cameras (when used with an app). They operate by emitting light into the skin and measuring the variations in light absorption due to blood volume changes with each heartbeat. The resulting PPG waveform is a pulsatile signal from which heart rate and rhythm can be inferred. The advantages of PPG for AF screening include: (1) Ubiquity and convenience—millions of people already wear PPG-equipped devices daily, providing a ready platform for passive monitoring; (2) Low cost and non-invasiveness—no electrodes or gel needed, and no significant user action required, so monitoring can be near-continuous and at scale; (3) 24/7 data collection—devices can collect data during normal activities and even sleep, increasing the chance of capturing short AF episodes that would be missed in a doctor's office [2].

The feasibility of this approach has been demonstrated by large studies. The Apple Heart Study (Perez et al., 2019) was one of the first and largest, showing that smartwatch PPG could identify irregular rhythms consistent with AF with a high specificity (the false-positive rate was low, as evidenced by an 84% PPV). Another trial by Huawei in China (Guo et al., 2019) similarly showed a PPV around 87–92% for their PPG algorithm. The recent Fitbit Heart Study (2021) improved specificity further (98% PPV) by incorporating strict criteria for an AF alert—requiring persistence of the irregular pulse for at least 30 minutes and only analyzing data when the user was still to avoid motion noise. These studies underscore the clinical promise: if PPG-based detection algorithms can be made accurate and reliable enough, they could enable widespread AF screening and early diagnosis, potentially preventing strokes in people who would otherwise go undiagnosed [3].

PPG signal characteristics and limitations: A PPG signal from the wrist or finger is essentially a pulse waveform corresponding to blood volume changes with each heartbeat. In sinus rhythm (normal heart rhythm), these pulses occur at fairly regular intervals, with slight variability due to respiration and autonomic tone. During atrial fibrillation, the ventricular contractions are irregularly timed; this manifests in the PPG as irregular intervals between pulse peaks and often varying pulse amplitudes (due to variable stroke volume when the heart rhythm is erratic). Thus, the primary signature of AF in a PPG recording is a highly irregular pulse interval time-series.

Many algorithms exploit this by calculating statistical measures of interval variability or randomness (for example, the standard deviation of beat-to-beat intervals or entropy measures) to detect AF. If the variability exceeds a certain threshold, the segment is classified as possible AF.

However, distinguishing AF from normal sinus rhythm or other arrhythmias using PPG is not always straightforward. A major reason is that PPG does not capture the electrical activity of the heart—crucially, it cannot show the absence of P-waves (atrial contractions) which is a definitive ECG hallmark of AF. PPG-based detection must rely on indirect evidence like irregularity and subtle changes in pulse morphology. This can lead to ambiguity in certain cases [4]. For example, consider atrial flutter with variable block: on ECG it has regular flutter waves, but irregular ventricular response; on PPG, it would simply look like an irregular pulse, potentially mimicking AF. Frequent premature beats (PACs or PVCs) are another common confounder: a bigeminal pattern of PACs (every other beat premature) produces an interval sequence like short–long–short–long, which has high variability but is not AF (the underlying rhythm is regularly irregular rather than chaotic). An algorithm that relies purely on statistical variability might misclassify such cases as AF. Indeed, some researchers explicitly add logic to detect PAC/PVC patterns (for instance by analyzing Poincaré plots of intervals for the distinctive clustering those patterns produce) to avoid false positives.

Another limitation is the noise susceptibility of PPG, especially from wrist devices. Motion artifacts are the foremost issue: movements of the hand or arm, or loose contact of the sensor on the skin, can introduce large fluctuations in the PPG signal unrelated to blood flow. These artifacts can produce false "peaks" or obscure true peaks. PPG is also affected by changes in peripheral circulation—for instance, vasoconstriction in cold weather reduces signal amplitude, and different skin pigmentation can affect the signal-to-noise ratio (darker skin absorbs more light, yielding a lower signal amplitude). Additionally, ambient light leakage, sweat, or sensor pressure changes can distort readings. A study by Paliakaitė et al. (2021) systematically characterized artifacts in wrist PPG recordings and found that depending on the activity, anywhere from 14% to 49% of the recording duration could be corrupted by noise, such as motion and device displacement artifacts. This underscores that any successful PPG-based AF detector must include robust artifact handling—either by cleaning the signal or by detecting low-quality segments and ignoring them [5].

In contrast, ECG recordings (especially from patch monitors or hospital telemetry) are more resilient in active patients, and even when noise occurs (e.g., muscle artifact), the ECG's direct measurement of electrical impulses often still allows identification of R-waves and the underlying rhythm. Thus, the performance gap between PPG and ECG is most pronounced in real-world conditions when the user is not perfectly still. Under ideal conditions (e.g., patient at rest, good perfusion), PPG can track heart rhythm quite accurately—and as we will discuss, algorithms have achieved high accuracy on clean PPG data. The challenge is maintaining that accuracy in everyday conditions. Our work is motivated by this challenge: to push PPG-based detection closer to the reliability of ECG by using advanced methods to combat noise and interpret the pulse pattern intelligently.

Existing approaches to PPG-based AF detection: Research over the past decade has produced a large body of work on algorithms for AF detection from PPG. Broadly, these can be categorized into three groups: (1) Classical signal processing and feature-based methods, (2) Machine learning classifiers using hand-crafted features, and (3) Deep learning approaches that learn features from the data. We review each in turn, as well as strategies used to improve performance (particularly with respect to noise robustness), in order to position our project in the context of state-of-the-art.

(1) Traditional feature-based methods: Early algorithms borrowed ideas from heart rate variability (HRV) analysis in ECG. A simple but effective approach is to detect pulses in the PPG, derive the sequence of pulse-to-pulse intervals (PPIs), and compute statistical metrics to quantify irregularity. Common features include the standard deviation of NN intervals (SDNN) and the root mean square of successive differences (RMSSD) between intervals. In AF, both of these tend to be elevated due to high variability. Entropy measures, such as sample entropy or approximate entropy, have also been used to capture the unpredictability of the interval time series—AF generally produces higher entropy compared to sinus rhythm. Threshold-based classifiers can then be applied: for example, if entropy and variability exceed predefined thresholds, label the segment as

AF [9].

One notable example, by Bashar et al. (2019), developed an AF detection algorithm for smartwatch PPG that combined two key features: normalized RMSSD and sample entropy of the pulse intervals. They found these features to be well-separated for AF vs normal rhythm, and instead of using two separate thresholds, they combined them into a single discriminant score. Additionally, they included a post-processing step to detect PAC/PVC patterns using Poincaré plot analysis and thereby reduce false positives due to frequent ectopic beats. In a test on 30-second PPG segments from a Samsung smartwatch dataset, their method achieved sensitivity around 95% and specificity ∼97% for AF detection [33]. This performance is comparable to many single-lead ECG algorithms. However, it was noted that they only analyzed segments deemed "clean" from motion artifact—an accelerometer-based filter was used to discard noisy segments. In practice, they could reach such high accuracy by effectively analyzing only high-quality data and ignoring periods of motion. This reflects a common strategy in PPG algorithms: ensure high precision by not attempting to classify when the data is unreliable (at the cost of some sensitivity if AF happens during those excluded periods).

Other classical features include frequency-domain metrics. For instance, some works compute the frequency spectrum of the beat interval series (tachogram) and use the absence of any dominant periodicity as a sign of AF (since sinus rhythm might show a peak corresponding to respiratory sinus arrhythmia around 0.1 Hz, whereas AF's interval sequence is more broadband). Poincaré plot features (SD1, SD2—measures of short-term and long-term variability) have also been used: AF typically yields a more circular Poincaré plot (high SD1 and SD2), whereas sinus rhythm yields an elongated shape (lower beat-to-beat variability vs overall variability). Some smartphone apps like FibriCheck employed a combination of such features with decision logic to achieve medical-grade accuracy in a controlled setting. In general, well-crafted feature approaches can deliver sensitivity and specificity in the 90–95% range when evaluated on relatively clean data or short segments in a supervised setting [34].

However, these hand-crafted methods have limitations: they often involve multiple heuristic thresholds that may need re-tuning for different devices or populations. They also might struggle if the signal deviates from assumptions (e.g., if peak detection misses beats, the computed features degrade). The advantage is that they are computationally light—important for implementation on low-power wearables—and are somewhat interpretable (e.g., a high entropy value clearly corresponds to irregularity, which aligns with clinical reasoning for AF). Indeed, some of the first commercially released algorithms (like the Apple Watch's irregular rhythm notification in 2018) were likely based on such feature logic augmented by safeguards to minimize false alarms. For example, Apple's algorithm required a series of irregular readings and only triggered when the user was inactive, in order to maximize specificity.

(2) Classical machine learning with HRV features: Moving beyond fixed thresholds, many researchers have applied standard machine learning classifiers to a rich set of features derived from PPG. In these approaches, one extracts a feature vector from each segment (which may include time-domain stats, frequency-domain measures, non-linear metrics, and sometimes signal quality indices), and then trains a classifier such as support vector machine (SVM), decision tree, or random forest to distinguish AF from normal (and sometimes other rhythms). The machine learning model can learn an optimal combination of features, potentially improving accuracy over any single threshold rule.

For example, Liao et al. (2022) studied how the length of the recording affects detection performance using a smartwatch PPG dataset. They extracted a variety of HRV features from segments of different lengths (10 beats, 25 beats, 40 beats, 80 beats) and used an SVM classifier. They found that using about 25 heartbeats (∼20–30 seconds of data) gave the best discrimination between AF and sinus, with an AUC (area under the ROC curve) of approximately 0.97. Using only 10 beats was notably less accurate (AUC ∼0.945), while extending beyond 25–40 beats yielded only marginal gains and could even degrade performance if the longer window included noise or transitions. This result reinforces the clinical notion that at least 20–30 seconds of data are needed to confidently diagnose AF (consistent with guidelines that AF should be present for at least 30 seconds). Liao et al. also explicitly included data with frequent premature beats in their testing and demonstrated that their classifier, trained on diverse rhythms, could still achieve high accuracy (∼95% sensitivity and specificity) by learning differences in features like regularity patterns that distinguish AF from, say, bigeminy [35].

Another study by Ramesh et al. (2021) took a hybrid approach: they used traditional time and frequency features from the PPG intervals but then fed those into a deep neural network (a 1-D convolutional neural net) to classify AF vs normal. Importantly, they leveraged transfer learning by first pre-training their model on a large ECG dataset for AF detection (where labels are plentiful) and then fine-tuning on a smaller PPG dataset. This improved the model's generalization, essentially using the fact that the "irregularity patterns" learned from ECG R–R intervals can also apply to PPG pulse intervals. Their approach exceeded 90% accuracy on their test set, illustrating the benefit of combining human-crafted features with machine learning and even deep learning elements [36].

Overall, classical ML classifiers can achieve excellent performance when trained and tested on the same type of data distribution. They offer more flexibility than hard thresholds and can incorporate many features simultaneously. However, they still rely on the quality of the feature engineering and do not inherently solve the noise problem—noisy inputs can lead to garbage features, which could mislead the classifier unless steps are taken to filter or flag low-quality data. Thus, many of these systems also include a preceding signal quality check: if a segment's quality is below some threshold (for example, low signal amplitude or a high noise index), the system may reject that segment or output an "indeterminate" result rather than risk a misclassification. This is considered good practice in medical-grade algorithms—it is often better to withhold a diagnosis than to give a false one based on poor data.

(3) Deep learning approaches: In recent years, end-to-end deep learning has been applied to AF detection from PPG with promising results. Deep learning models can automatically learn complex features from the raw waveform or derived sequences, potentially capturing subtle patterns that manual features might miss. Several types of neural network architectures have been explored:

- **Convolutional Neural Networks (CNNs) on raw PPG waveform:** A CNN can learn shape-based features and local patterns in the PPG signal that correlate with AF. For instance, Tison et al. (2018) developed a deep CNN for AF detection using the Apple Watch PPG data [39]. Their model effectively learned to recognize the irregular pulse pattern amidst other physiological variations. Reportedly, the neural network approach yielded sensitivity around 98% for detecting AF, with specificity $\sim$90% in a validation dataset. Another work by Yoon et al. (2021) used a deep CNN to classify segments into AF, normal, or other arrhythmias, achieving over 94% accuracy in a controlled evaluation. These models typically require a considerable amount of training data and computational power; in practical wearable deployments, the model might run on a server or be simplified to run on-device.
- **Recurrent Neural Networks (RNNs) or Transformers on pulse interval sequences:** Instead of feeding the entire high-frequency waveform, an alternative is to feed the sequence of detected beat intervals (and possibly their amplitudes or other beat-level features) into a sequential model. LSTM (Long Short-Term Memory) networks or Transformer models can then process this sequence to detect irregularity patterns characteristic of AF. The advantage is a drastic reduction in input data (e.g., 30 seconds of PPG at 100 Hz is 3000 samples, whereas 30 seconds might contain $\sim$30 beats, so just 30 interval values) and a focus on the timing information. Kudo et al. (2023) recently compared several input representations and models for AF detection. They found that using the first-order differences of the inter-beat intervals as input to a small CNN+Transformer hybrid model gave the best results, with an F1-score of 0.80 for AF vs normal vs other arrhythmia classification [23]. The CNN part captured short-term patterns in the interval sequence (like sudden changes), while the Transformer part captured longer-term dependencies and overall irregularity structure. Notably, they included a third class for "other arrhythmias" to help the model learn the distinction between truly random AF and rhythms with structured irregularity (like frequent PACs). This reflects a growing trend to move beyond binary classification and explicitly account for confounding rhythms in model training.
- **Hybrid models with multiple inputs:** Some researchers have combined inputs, such as using both the raw PPG and the interval sequence, or using multi-lead PPG (if available). The idea is to provide the model with richer information—e.g., the morphology of each pulse and the timing between pulses. A hybrid model might have a CNN branch analyzing the

waveform and an RNN branch analyzing intervals, merging the learned features before the final decision. This can potentially improve robustness: the timing analysis might be more resistant to certain noise (like small baseline shifts), while the waveform analysis might catch things like alternating pulse amplitude patterns that intervals alone wouldn't show. So far, literature in PPG AF detection has been dominated by either one or the other, but such multi-branch models could be an interesting direction (and we plan to explore a form of this in our project).

Deep learning models have achieved performance on par with classical approaches in many validation studies, often reporting sensitivity and specificity in the 90–95% range. An advantage is that, given enough training data, they can potentially generalize better and require less manual tuning of parameters. They can also incorporate more complex criteria. For example, a deep model might implicitly learn to ignore segments with too much noise (if those segments are labeled as uncertain or non-AF in training). Some advanced works have introduced attention mechanisms or self-supervised learning to cope with noise. Ding et al. (2024) proposed a "signal-quality-aware" training framework, where a model is trained on multi-source data with a self-distillation approach so that it learns from high-quality examples how to handle low-quality ones. Their model, when applied to a challenging ICU dataset (MIMIC, which has simultaneous PPG and ECG), outperformed baseline CNNs by a large margin, achieving about 88% accuracy (83% sensitivity, 90% specificity) on AF detection. This was a notable improvement over earlier attempts on the same data (which had ∼73% accuracy) and narrowed the gap to ECG performance in that setting [38].

Another interesting development is transfer learning between ECG and PPG. Since AF detection from ECG is well-studied, some researchers use ECG data to pre-train models or to guide PPG model training. A 2023 study introduced a "Siamese" network (SiamAF) that learns a shared representation for simultaneous PPG and ECG inputs, effectively using the ECG as a teacher for the PPG feature extractor. While our project focuses purely on PPG, such approaches highlight the creative strategies being explored to maximize PPG accuracy [37].

In summary, deep learning holds promise for pushing PPG-based detection closer to ECG-level performance, especially as more training data becomes available from large wearables studies. But with that said, even the best algorithms will falter if the input signal is too corrupted. Therefore, much of the recent research also emphasizes noise reduction and artifact handling, which we discuss next.

Noise artifact management in PPG AF detection: A recurring theme is that handling noise is crucial for any PPG-based method to work reliably in practice. Various strategies have been employed:

- **Signal preprocessing and filtering:** Most pipelines start with band-pass filtering the PPG (typically passing 0.5–5 Hz) to remove slow drifts and high-frequency noise. Some use more sophisticated adaptive filters or wavelet denoising to target motion artifact signatures. For example, accelerometer data (which measures motion) can be used in an adaptive filter to subtract motion-induced components from the PPG signal. In our literature survey, nearly every method applied some form of preprocessing to improve the signal quality before analysis.
- **Signal Quality Indices (SQI):** Researchers have defined metrics to quantify PPG signal quality. Simple SQIs include the percentage of time the signal is above a certain amplitude (low values might indicate poor contact), or the ratio of energy in the pulse band vs total energy (a lot of energy outside the heart rate band might mean noise). More complex SQIs involve template matching—correlating the PPG waveform to an expected pulse shape and flagging low correlations as noise. Some works combine multiple criteria into an aggregate quality score [40]. In the context of AF detection, SQIs are often used to exclude or down-weight bad segments. For instance, an algorithm might decide: "if SQI $<$ 0.5 for this 30s window, do not attempt classification and instead wait for a cleaner segment."
- **Motion artifact detection:** When accelerometer data is available (as in most smartwatches), many systems explicitly detect motion and either pause analysis or use sensor fusion. Some academic prototypes fuse PPG and accelerometer data through machine learning to distinguish true heart rate changes from motion-induced changes. An example is RhythmiNet (2020), which took PPG and ACC signals together into a deep network and

improved AF detection during exercise by accounting for the motion context [41]. While our project will focus on PPG alone (to simulate scenarios where only PPG is used), we acknowledge that in practice, a commercial solution would likely leverage the accelerometer. Indeed, as mentioned, Fitbit's algorithm only runs during "stationary" periods determined by the accelerometer. In our approach, we emulate this by using PPG-intrinsic measures to gauge when a segment is stationary enough.

- **Artifact simulation and augmentation:** A novel approach to improving robustness is to simulate realistic artifacts and use them in training. Paliakaitė et al. (2021) not only analyzed artifacts but also created a model to generate them, allowing testing of how detection algorithms perform with known noise injections. By augmenting training data with simulated motion artifacts, one can teach a model to either recognize and ignore those patterns or at least not be overconfident during noisy periods. We plan to leverage such ideas by adding artificial noise to some training examples (e.g., adding bursts of high-frequency noise or baseline wander) to make our model more tolerant [42].
- **Two-stage or cascaded algorithms:** Some deployed systems use a two-step decision: first check signal quality, then check rhythm. Others go further to include a confirmation stage: e.g., if a 30s segment is classified as AF, the system might require it to be repeated multiple times (persist for several minutes) before declaring an AF event, as a safeguard. While this improves reliability, it can delay detection. This approach was evident in the wearable trials (Apple required multiple irregular readings and Fitbit required 30 min of continuous irregularity). In our evaluation, we will consider similar logic when comparing to ECG: e.g., counting how many consecutive segments of AF are needed before we'd "alert."

Despite significant progress, fully closing the gap to ECG is still an open problem. The best PPG algorithms under research conditions report performance close to ECG-based detectors (often >95% accuracy on curated datasets). In real-world deployment, however, performance tends to drop due to the variability of conditions. The Fitbit study's sensitivity of ∼68% for AF episodes, for instance, is a reminder that even a very specific algorithm can miss episodes if they occur during activity or are short. Our project is essentially targeting that space: improving the sensitivity of PPG detection (catching more AF) without sacrificing the high specificity (avoiding false alarms) that is needed for user and physician trust. We believe that by combining advanced algorithms (like deep learning for pattern recognition) with rigorous handling of signal quality (preprocessing, gating), we can significantly narrow the performance gap.

In the remainder of this report, we outline our methodology for building such a system and how we plan to evaluate it. We ground our approach in the lessons from the literature: use enough data (segment length) to capture AF's signature, incorporate noise checks, and utilize modern ML techniques that have shown success in related works. Ultimately, we will quantify how close we get to the "ECG-level" benchmark and identify what gaps remain.

Chapter 3

Project Plan and Methodology

We propose a PPG-based AF detection pipeline that integrates robust preprocessing, signal quality assessment, and a machine learning classifier (potentially a hybrid deep learning model) trained to differentiate AF from normal rhythm (and other confounding patterns). Our aim is to optimize this pipeline so that on test data it achieves high sensitivity and specificity, comparable to what an ECG-based algorithm would achieve on the same data. To ensure a fair benchmark, we will use datasets where both PPG and reference ECG are available; ECG will provide the ground truth of AF vs non-AF, but it will not be used as an input to the model. The project can be broken down into several components:

1. **Data acquisition and preparation:** We will use two main sources of data: (a) a public dataset of clinical recordings that include simultaneous PPG and ECG with AF annotations, and (b) a synthetic PPG data generator for augmenting our training set. Specifically, we have obtained the MIMIC-PERform Atrial Fibrillation Dataset, which contains 20-minute segments from 35 ICU patients (19 with AF, 16 in normal rhythm) with both ECG and PPG waveforms recorded. The ECG has been expert-labeled for AF episodes, so we can use that as ground truth to label corresponding PPG segments [13]. This dataset provides real-world signals with true AF events and various noise typical of ICU (though ICU PPG might be higher quality than wearable PPG, it still has artifacts from patient movement or poor perfusion). However, 35 subjects is relatively small for training a complex model, so we will expand our training data using a PhysioNet PPG simulator. The simulator by Solosenko et al. can generate realistic PPG signals and can simulate arrhythmias like AF as well as different noise conditions [14]. We plan to generate a large number of synthetic PPG segments (possibly thousands) with known rhythm labels (AF or normal), under various noise scenarios (clean, mild motion, heavy motion, etc.). These will be used to pre-train our model and to augment the variability of training examples. By fine-tuning on the real dataset after pre-training on synthetic data, we hope to capture real-signal nuances while benefiting from the broader training of the model on diverse scenarios (a form of transfer learning as used in prior works).

   We will partition the real dataset into training, validation, and testing sets by patient (to ensure that we test generalization to unseen individuals). Tentatively, about 25 subjects will be used for training, 5 for validation (parameter tuning), and 5 for final testing. This yields on the order of a few hundred 30-second segments in each set (since 20 min per subject can produce up to 40 segments of 30s, though AF subjects will have some fraction of those labeled AF and others normal if AF was intermittent). The synthetic data generation will not precisely mimic the real data distribution, but it will provide additional AF examples to prevent overfitting given the limited real AF cases.

2. **Signal preprocessing module:** We are implementing a preprocessing pipeline to filter and normalize the PPG signals and extract basic features like pulse peaks. The steps include [15]:

   (a) **Band-pass filtering:** Each PPG signal is filtered with a band-pass (approximately 0.5–8 Hz). This removes baseline wander and motion drift (below $\sim$0.5 Hz) and high-frequency noise (above ∼8 Hz, which is beyond the frequency of any physiological pulse changes). The upper cutoff of 8 Hz covers heart rates up to about 480 bpm, which is far above any realistic heart rate—so essentially we retain all heart rate information but drop high-frequency noise that could come from sudden movements or sensor issues.

(b) **Resampling/normalization:** We will resample signals to a uniform sampling rate (e.g., 50 Hz) if they are not already uniform, because the ICU data might be 125 Hz and our synthetic might be at a different rate. We will also scale each segment's amplitude (e.g., z-score normalization or min-max scaling) so that amplitude differences between individuals (due to wrist size, skin tone, etc.) do not directly bias the model. Instead, the model can focus on pattern of the waveform rather than absolute values.

(c) **Peak detection:** A reliable peak detection algorithm will be applied to identify the timing of each pulse. We are initially using an established method by Elgendi et al. (2013) for PPG peak detection, which uses slope and amplitude criteria to find systolic peaks [16]. We have tested this on a few sample signals to ensure it finds all true beats (comparing against ECG R-peaks for verification in the MIMIC data). Accurate peak detection is vital, as it underpins the calculation of the inter-beat intervals. If peak detection fails during a noisy stretch, that stretch would likely be flagged as low-quality.

(d) **Signal segmentation:** We will analyze data in fixed-length segments (windows) for classification. Based on literature and practical considerations, we choose a 30-second window (with tolerance for slightly shorter if needed to align with heartbeats). This aligns with clinical definition (AF episodes lasting ≥30s are considered significant) and was supported by Liao et al.'s finding that 25–30 beats (∼30s) is an optimal length for analysis [17]. For each 30s segment, we will derive its label from the ECG ground truth (in the training set we know exactly which periods are AF). For continuous signals where AF may start or stop partway through, segments that straddle a transition might be excluded or labeled based on majority. In MIMIC, many recordings might be entirely AF or entirely normal for the 20 min; if so, all segments from an AF recording would be labeled AF. In any case, segmentation will ensure we present the classifier with manageable chunks rather than an entire long recording. We will likely use overlapping segments during inference (e.g., update every 10s) but for training each segment is a separate sample.

3. **Signal Quality assessment:** In parallel, we will compute one or more signal quality indices for each segment. Our current plan is to use a combination of simple SQIs: (a) the ratio of the PPG's energy in the heart rate band (0.5–3 Hz) to total energy (an indicator of whether a dominant pulsatile component is present), (b) the percentage of detected beats that are "reasonable" (we expect roughly 30–40 beats in 30s; if peak detector finds say 100 or 5, something is wrong), and (c) a morphological correlation—we can take an average pulse shape in the segment and measure how similar each beat is to this template; low correlation could indicate artifact-distorted pulses [18]. We will empirically determine a threshold on these metrics using the training data (which has ECG to confirm if a segment was likely noise—e.g., if ECG shows a steady heart rate but PPG peaks are wildly irregular, that indicates PPG noise). During model training, we might choose to exclude segments below a certain quality, or label them separately as "noise" to possibly train the model to recognize and ignore noise. During deployment (or evaluation), we will implement a gating: if a segment's SQI is below threshold, the system will output "signal too noisy" rather than AF or Normal [19]. This strategy is intended to keep false positives low by not guessing on poor data. The downside is some sensitivity loss (we might miss AF that occurs only during very noisy periods), but the trade-off is necessary for reliability. We will document how often this gating occurs, as it effectively defines time periods where the wearable "can't tell"—ideally this should be a small fraction of total monitoring time.

4. **Feature extraction and modeling:** We will explore two parallel modeling strategies and then possibly a hybrid: a traditional feature-based classifier and an end-to-end learned model.

(a) **Feature-based classifier:** We will compute a set of interpretable features from each segment's PPG and PPI sequence, and train a classifier on those features. These features may include: heart rate mean and standard deviation, RMSSD of intervals, sample entropy of intervals, pNN50 (percentage of interval differences >50 ms), and coefficient of variation of pulse amplitude [20]. We'll also include features to help spot non-AF irregularity, e.g., regularity metrics that might distinguish bigeminy from AF (such as autocorrelation of the interval series—PAC bigeminy would show a strong alternating pattern, whereas AF would not). Another could be the skewness or kurtosis of the distribution of intervals (AF's interval histogram is often broader and more symmetric, while sinus rhythm with occasional ectopics might show two distinct clusters of intervals). Additionally, we might incorporate the signal quality itself as a feature, so the classifier learns to be cautious (or even automatically classify "noise" if quality is extremely low). Once these features are computed, we will train a classifier like a Random Forest or an SVM [21]. Random Forests are appealing for their robustness and ability to rank feature importance, whereas SVMs with an RBF kernel often perform well in binary classification of HRV data. We will use the validation set to prevent overfitting and choose hyperparameters. The feature-based approach will serve as a baseline—it might not reach the performance of a more complex model, but it provides interpretability and a sanity check (if the ML approach fails, at least a reliable feature method can still be used as a fall-back).

(b) **Deep learning model:** In parallel, we are developing a deep learning model that can take either the raw (or filtered) PPG segment, or a derived representation (like the interval sequence), and directly learn to classify AF vs normal. One architecture we plan to try is a 1-D CNN followed by a sequence model. For instance, a CNN could first process the PPG waveform to extract local features (like detecting individual pulse peaks, shapes, etc.), then an LSTM or Transformer could process the sequence of CNN outputs over time to make the final classification [22]. This is somewhat analogous to how a human might look at both the pattern of intervals and the morphology of pulses. An alternative input to the model is the sequence of intervals themselves (a much lower-dimensional input). In that case, we might use a small CNN to capture local differences in successive intervals and a Transformer encoder to capture long-range irregularity. Transformers are suitable due to their ability to model long dependencies and have been used in recent AF detection tasks with good results. Our architecture will likely be influenced by Kudo et al. (2023) who successfully combined CNN and Transformer on interval differences. We will however simplify as needed to avoid overfitting given our data size—possibly using fewer layers or lower-dimensional embeddings [23].

A potential architecture we are considering is: CNN feature extractor + Transformer classifier. The CNN (with maybe 2–3 convolutional layers) would take as input a time-series (either the PPG or an interval time-series expanded to a fixed length vector) and output a feature map. The Transformer encoder (with maybe 1–2 self-attention layers) would then treat this feature map as a sequence and compute contextual representations, which finally go into a dense layer for classification [24]. We will incorporate regularization such as dropout and perhaps perform data augmentation on the fly (e.g., jittering the input slightly or adding noise) to improve generalization. The model will output a probability (or confidence) of the segment being AF, and we will apply a threshold (probably 0.5) to decide AF vs not. During training, we will use a binary cross-entropy loss for classification. If we decide to include a "noise" class, it would become a 3-class problem with a softmax output.

We have not finalized whether to use the raw waveform or the interval sequence as primary input—we will experiment with both using the validation set for guidance. Using raw waveforms might capture more information (like pulse amplitude variations), but it also introduces more noise and increases model complexity. Using intervals (plus maybe a few aggregate features) drastically simplifies the input and focuses the problem on irregularity detection, which might yield a more robust result given limited training data. A compromise could be to use interval-based input for the main classification, supplemented by a few additional inputs like average pulse amplitude or variability in pulse shape (to hint the model about things like alternating pulse strengths in AF). This can be done by feeding those as additional features concatenated with the network's learned features before the final layer.

5. **Training strategy:** As mentioned, we plan to use synthetic data to pre-train the deep learning model. Concretely, we can generate, say, 10,000 synthetic 30-second segments (half AF, half normal, with a variety of heart rates and with random injection of motion artifact noise in some). We train the model on this synthetic dataset for a number of epochs until it learns a general concept of AF (which is basically irregular intervals). Then we fine-tune on the real training data (from MIMIC) for a few more epochs, which adjusts the model to the real signal characteristics [25]. We will monitor performance on the validation set to choose the epoch at which to stop (using early stopping if validation loss starts to increase).

   The feature-based model does not need pre-training but we will use cross-validation on training data to ensure its parameters (or feature selection) are well-tuned. If the feature model performs exceedingly well, it might influence the design of the deep model (for instance, if we find sample entropy is the most powerful feature, we might ensure our deep model's architecture is sensitive to similar patterns).

   We will also incorporate data augmentation during training of the deep model. Besides the synthetic data (which is a form of augmentation), we can augment real PPG segments by slightly warping them (e.g., simulating a slight heart rate change by time-stretching the signal, or adding mild Gaussian noise, or randomly scaling amplitude). The goal is to expose the model to slight variations and noise so it doesn't become too brittle. We will be careful not to distort the rhythm label (e.g., we wouldn't add ectopic beats to a "normal" segment without changing its label, as that would effectively turn it into an AF-like irregular segment). Augmentations will be label-preserving in nature (small enough not to change AF vs not). For example, for an AF segment, adding a bit of baseline wander or a few random dropouts in the waveform can teach the model to still recognize irregularity in presence of such imperfections.

   As part of training, we will also experiment with a small portion of the model dedicated to signal quality prediction—for instance, adding an auxiliary output that predicts whether a segment is high-quality or not [26]. This multi-task learning could help the model learn to internally distinguish noise from arrhythmic patterns. However, given our relatively small dataset, we might keep it simple and handle quality outside the model.

   We have also identified fallback options in case certain components prove too challenging: for instance, if the deep learning approach does not yield a clear improvement over the feature-based method, our fallback is to optimize the feature model (maybe using a more complex ensemble or fine-tuned thresholds) and present that as the main result. The deep model, even if not outperforming, would still be discussed with analysis of why it might have struggled (e.g., data quantity issues). We also have a contingency that if the MIMIC dataset is too limited, we could leverage other public datasets (like the 2015 Challenge dataset of PPG during arrhythmia, or any new wearables data that might be available) to supplement test validation. So far, this looks unnecessary, but it's a backup plan.

   To demonstrate real-world feasibility beyond offline evaluation, we will implement a complete hardware-integrated AF detection system using an August International smartwatch as the sensing platform. The deployed system will process streaming PPG and accelerometer (IMU) signals in real time, executing the full pipeline—data acquisition, preprocessing, signal-quality estimation, windowing, inference, and decision logic—under realistic compute and energy constraints. Importantly, the target deployment setting assumes that inference should be executable on-device (i.e., directly on a smartwatch-class platform) wherever possible, rather than relying on server-side computation, which imposes strict constraints on memory footprint, latency, and energy consumption. This assumption is consistent with recent industrial edge-AI workflows that emphasize compact, CPU-runnable models tailored to resource-constrained SoCs, where models are generated and optimized to run directly on-device under tight power and memory budgets [29].

   The August International smartwatch will be used to collect synchronized PPG and triaxial accelerometer signals during continuous wear. Sensor streams will be timestamped and aligned to a common clock to ensure that motion features and PPG-derived rhythm features are computed over identical windows. We will implement a real-time buffering mechanism (ring buffer) that maintains the most recent T seconds of sensor data (e.g., 60 s), enabling sliding-window inference. We will integrate the smartwatch into a runnable inference stack via one of the following deployment topologies (both are hardware-integrated and preserve real-time operation):

- **On-watch inference (preferred when supported):** the trained model will be exported to a wearable-compatible format and executed directly on the watch. This configuration provides true edge inference and allows direct measurement of end-to-end latency and power cost on the wearable. In line with industrial "direct-to-device" deployment patterns (e.g., Neuton AI compact CPU-run models), this pathway treats on-device executability as a first-class design constraint when selecting input representations, model size, and inference cadence [27,30].
- **Watch-to-edge inference (robust implementation path):** if the watch environment limits direct model execution, the watch will stream raw or lightly preprocessed PPG/IMU windows to a paired edge runtime (e.g., a companion device) over a low-latency link (e.g., BLE). Inference will still run in real time, and the pipeline remains identical; the only difference is the execution location of the classifier. This ensures the project delivers a complete, operational hardware pipeline even under restricted smartwatch programmability. Even in this configuration, we will preserve the same compute-budget assumptions by constraining the model to remain wearable-class (small memory footprint and low compute), so that results remain representative of eventual on-watch deployment [28].

In both cases, the watch remains the primary sensing hardware, and the system is evaluated under continuous streaming conditions rather than offline file-based inference. We will implement lightweight, streaming-friendly preprocessing compatible with embedded constraints: band-pass filtering of PPG to remove baseline drift and high-frequency noise using computationally efficient IIR/FIR filtering with persistent state (streaming operation rather than per-window recomputation); normalization per window (e.g., z-score or robust scaling) to reduce inter-subject amplitude variability and sensor-contact effects; and peak detection and PPI extraction (for interval-based variants), implemented with thresholding + refractory logic to ensure stable beat detection on device.

Signal quality will be estimated using both PPG-intrinsic metrics and IMU-derived motion features. Specifically, we will compute: (i) spectral concentration in the physiological heart-rate band, (ii) beat-count plausibility and refractory violations, (iii) morphology consistency across beats, and (iv) an IMU motion index (e.g., accelerometer magnitude variance/activity). These metrics will be used either as a gating mechanism (suppressing decisions for low-quality windows) and/or as auxiliary inputs to a noise-aware classifier, explicitly reducing motion-driven false positives. Inference will be executed on fixed-length windows (e.g., 30 s) with periodic updates (e.g., every 5–10 s). The classifier outputs a probability $p(\mathrm{AF})$ per window, which is converted into a robust decision using temporal smoothing rules (e.g., requiring K consecutive AF-positive windows, and sufficient SQI) to avoid sporadic misclassification from transient artifacts [31]. When SQI falls below threshold, the system will output an "unreliable signal" state rather than forcing an AF/Normal classification. Because on-device deployment is a core requirement, we will treat model complexity as a constrained resource and prioritize architectures and inputs that minimize memory and compute while preserving performance under motion and noise. We will optimize the deployed model for wearable constraints using compression techniques such as quantization and architecture simplification to reduce memory footprint and inference cost. We will then benchmark deployment-level metrics including: (i) end-to-end latency per inference cycle (preprocessing + SQI + inference + post-processing), (ii) sustainable update rate, (iii) memory footprint, and (iv) energy/battery impact during continuous monitoring. Finally, we will validate numerical consistency between offline and deployed inference (within tolerance) and report performance stratified by motion level to quantify robustness in realistic wearable conditions.

Chapter 4

Evaluation Plan

To determine whether our PPG-based AF detection system "approaches ECG-level accuracy," we will conduct a rigorous evaluation on a held-out test set and compare performance against reference standards. Our evaluation strategy includes the following components.

- **Primary classification metrics:** We will compute sensitivity, specificity, accuracy, precision (PPV), and F1-score for AF detection on a per-segment basis. Sensitivity (recall) is the proportion of true AF segments correctly identified by our algorithm, while specificity is the proportion of normal segments correctly identified as normal. We target high values for both (ideally $> 90\%$). The F1-score provides a single measure that balances precision and recall, which is useful if the dataset is class-imbalanced. For context, ECG-based AF detectors often achieve sensitivity and specificity in the high-90% range in clinical settings; our aim is to be not far below that. We will also plot the ROC curve by varying the decision threshold on the model's output probability and report the AUC; values near 1.0 indicate excellent discrimination.

- **Segment-level vs episode-level evaluation:** Because AF is clinically discussed in terms of episodes (continuous AF duration), we will translate segment-level predictions into episode detection performance. Using ECG-derived onset/offset times, we will simulate alert logic (e.g., requiring at least two consecutive 30 s segments predicted as AF to trigger an episode detection). We will report: (i) episode sensitivity (fraction of true AF episodes detected at least once), (ii) episode false alarm rate (how often AF is flagged when none is present), and (iii) detection latency (time from AF onset to first alert), ideally within the first 30–60 s.

- **Comparison to an ECG "upper bound" benchmark:** Since ECG is available in our dataset, we will also run an ECG-based AF detection baseline as a benchmark. This may be a simple algorithm using R–R irregularity and (where applicable) evidence consistent with P-wave absence, or alternatively the provided expert labels may be treated as the gold standard. We will quantify the gap between our PPG-based performance and this ECG reference. Any residual difference (e.g., 95% vs near-perfect ECG performance) will be discussed as the expected performance gap between PPG-only and ECG-based detection.

- **Robustness checks (signal quality and stress testing):** We will analyze performance as a function of signal quality. For the test set, we will stratify segments by SQI (or other noise proxies) and quantify whether performance degrades in low-quality conditions (e.g., sensitivity/specificity versus SQI). We will also run targeted stress tests using synthetic data: for example, injecting increasing noise into normal and AF segments to assess whether the model (i) abstains appropriately ("unreliable signal"), (ii) avoids dangerously wrong predictions, and (iii) retains AF detection under challenging but plausible noise levels.

- **False positive analysis:** Because a screening tool must maintain a low false-positive rate to preserve user and clinician trust, we will report specificity and PPV and also manually inspect false-positive cases. By reviewing corresponding ECG segments, we will categorize likely causes (e.g., ectopy such as PACs/PVCs, bigeminy-like patterns, or motion artifacts). This analysis will inform potential improvements (e.g., post-processing rules, refined SQI gating, or targeted augmentation) and will be discussed explicitly in the evaluation.

- **Benchmarking against requirements ("how close to ECG-level"):** We will quantify how close the PPG approach is to ECG-level detection. If ECG is treated as nearly perfect on these labels, then a gap such as 92% sensitivity vs 99% is the remaining performance difference. We may also report agreement measures such as Cohen's κ to summarize classification agreement beyond chance. We expect some gap, but if it is small, we can argue the system is approaching ECG performance; if large, this will be reported as an important limitation.
- **Computational performance and deployment practicality:** Although accuracy is the primary focus, we will also report deployment-relevant metrics including model size, estimated inference cost, and end-to-end runtime (preprocessing + SQI + inference + post-processing). If needed, we will propose optimizations such as model compression or quantization to ensure feasibility on wearable-class devices.

Success criteria. We consider the approach successful if we achieve $> 90\%$ sensitivity and specificity on the held-out test set at the segment level, with PPV also exceeding 90% under roughly balanced test conditions. To support the claim of "approaching ECG-level," we target performance within approximately 5 percentage points of an ECG benchmark (treated as near-perfect on labeled data). In addition, we aim for a sufficiently low false alarm rate such that continuous monitoring would not yield frequent spurious alerts. For example, 95% specificity implies 5% of normal segments may be falsely flagged, which could translate into multiple false alerts per day without additional logic. Therefore, we will also report "false positives per hour of recording" (or a similar rate-based metric), and we expect SQI-based gating to substantially reduce false positives at the cost of some abstentions ("no-calls") [32]

Coverage and abstention. We will evaluate coverage: the fraction of time windows for which the algorithm produces a definitive output versus abstaining due to low signal quality. If coverage is too low (e.g., skipping 50% of windows), high accuracy on the remaining windows may not translate into practical usefulness. We will therefore tune SQI thresholds to balance coverage and reliability, potentially using adaptive thresholds (e.g., stricter during high activity and more lenient during rest).

Protocol and reporting. All final evaluation experiments will be conducted on the held-out test set that the model has never seen to ensure unbiased assessment. The validation set will be used during development for model selection and threshold tuning, but final reported numbers will be test-set results. Finally, we will include qualitative examples (figures) illustrating typical behavior: an AF segment with a confident detection, and a noisy segment showing abstention or stable normal prediction. These visual examples complement quantitative metrics by demonstrating model behavior under realistic conditions.

Chapter 5

Ethical, Legal, and Societal Considerations

Our project primarily involves data analysis and algorithm development using existing datasets, so the direct ethical concerns are limited. However, there are a few points to address:

- **Use of medical data and privacy:** We are using the MIMIC-III Waveform Database (via the MIMIC-PERform subset), which is a publicly available resource of de-identified ICU patient data. This dataset has been rigorously de-identified to remove protected health information, and its use for research is covered under a data use agreement. We have completed the required training (the CITI "Data or Specimens Only Research" course) to access MIMIC data. Thus, our use of this data is in line with ethical guidelines and IRB approvals already in place for MIMIC. No new patient data is being collected by us, and no identifiable information is present. We ensure that any results we present (e.g., example waveforms) cannot be traced to an individual and contain no personal metadata.

- **Bias and generalizability:** One societal consideration is whether the algorithm performs equitably across different demographics. PPG signals can be affected by skin tone (darker skin may reduce signal amplitude) and other physiological differences. If our training data (ICU patients, mostly adults in critical care) is not representative of the general population (including younger wearable users or different ethnic groups), the algorithm might not generalize well. This is a limitation of many medical AI systems: biases in the training data can lead to lower accuracy for underrepresented groups. While our interim testing is on the same dataset, in the future, validating on more diverse data would be important. Ethically, one should ensure a screening tool is fair and does not underserve a particular group. In our report, we will acknowledge this and suggest that further training data from broad sources (including healthy wearable users) be used to mitigate any bias.

- **False alarms and psychological impact:** An AF alert from a wearable can cause anxiety for users, especially if it turns out to be false. In the Apple Heart Study, for example, a small fraction of users received alerts and many of them had no AF on follow-up, which could cause unnecessary worry or medical visits. Our project is mindful of this: we prioritize specificity and incorporate a "no-call" option for low-quality data to avoid spurious results. Ethically, designing the algorithm with a high bar for alerts aligns with the principle of "do no harm": we prefer to miss a short episode (which might be caught later) than to subject someone to repeated false alarms that could lead to stress or unnecessary anticoagulant treatment. We will discuss this trade-off in our evaluation. If implemented, clear communication to users about what an alert means and does not mean is crucial (e.g., "this is not a diagnosis, but a suggestion to get ECG confirmation").

- **Integration into clinical workflow:** From a professional standpoint, if such an algorithm were deployed, healthcare providers must decide how to handle it. Current guidelines typically require ECG confirmation of AF before starting treatment. Our project does not aim to change that, but to serve as a screening tool. Therefore, one ethical aspect is ensuring this remains a support tool rather than a standalone diagnostic that could bypass clinicians. In our documentation, we will reinforce that ECG is the gold standard and our PPG method is intended to assist in identifying candidates for ECG testing.

- **Regulatory compliance:** Any medical-related software, especially one that can affect patient management, would need regulatory approval (e.g., FDA, CE marking). This is beyond our project scope, but if taken forward, compliance with medical device software standards would be required, including risk assessments and demonstrating safety/effectiveness in clinical trials. Our current work is in a research context. We have implicitly considered safety by aiming for few false positives and by planning to label uncertain outputs rather than making potentially wrong calls.
- **Ethics of algorithm development:** We completed an internal ethical checklist with our supervisor. One question was whether our project involves human subjects or could cause harm. Since we use retrospective, anonymized data and do not interact with patients, there is no direct human subject research requiring additional approval. Another question concerns conflicts of interest: we are not funded by or affiliated with any company in this domain, so no conflict is present.

In conclusion, our project does not pose significant ethical risks. The main points are ensuring responsible use of data and addressing the implications of the algorithm's performance on users. We have addressed data privacy by using approved open data, and we incorporate fail-safes (e.g., not outputting a diagnosis on low-quality data) to align with patient safety. We will maintain honesty in communicating limitations, which is an ethical responsibility in reporting scientific results. For example, if our algorithm still misses certain cases, we will discuss this openly so others (or future clinicians considering such tools) understand what it can and cannot do.

Chapter 6

Future Timeline

We have defined a structured implementation plan aligned with the final project report deadline (Friday 12 June 2025, 13:00), focusing on a complete PPG-based AF detection pipeline that is robust to real-world noise and designed under wearable-class (on-device) compute constraints. In the first phase (January), we will finalize the end-to-end data pipeline and preprocessing stack. This includes confirming patient-wise train/validation/test splits on the MIMIC-PERform dataset, fixing the windowing protocol (30 s segments, with explicit handling of rhythm transitions), and completing a streaming-friendly preprocessing module consisting of band-pass filtering (0.5–8 Hz), resampling to a uniform rate (e.g., 50 Hz), and per-window normalization. In parallel, we will implement a first version of signal quality assessment based on simple, computationally efficient indices (e.g., physiological-band spectral concentration, beat-count plausibility, and morphology consistency), and define an explicit "unreliable signal" gating policy for low-quality windows. By the end of January, we will also establish a baseline feature-based classifier (e.g., Random Forest or SVM) using interval- and morphology-derived features, and obtain an initial validation benchmark to guide subsequent model development. Finally, we will complete the synthetic PPG generation pipeline and begin producing the large synthetic corpus required for deep model pre-training.

In the second phase (February), we will focus on deep learning model development under strict model size and inference-cost constraints, prioritizing architectures that remain feasible for wearable deployment. We will evaluate two primary input representations: (i) a compact interval/PPI-driven representation that directly targets rhythm irregularity with minimal dimensionality, and (ii) a filtered raw-PPG representation for cases where waveform morphology provides additional discriminative value. Model exploration will proceed iteratively across lightweight 1D CNNs and small hybrid sequence models (e.g., CNN–LSTM or CNN–Transformer-lite variants), with hyperparameters selected on the validation set. During this period, we will operationalize the synthetic pre-training and real-data fine-tuning strategy, including early stopping and label-preserving augmentation, and integrate the signal-quality gating mechanism directly into the evaluation pipeline. By the end of February, we aim to identify a small set of candidate deep models that reliably outperform the baseline on validation while meeting the targeted memory and compute budgets, and we will report performance both with and without SQI-based gating to quantify its impact on false positives and abstention rate.

In the third phase (March), we will conduct a thorough held-out test evaluation on unseen subjects and perform structured error analysis. We will report sensitivity, specificity, ROC-AUC (and PR-AUC where appropriate), alongside confusion matrices and performance stratified by signal quality and motion level (where motion proxies are available). We will then analyze failure modes to determine whether false negatives cluster in high-noise windows, whether specific confounding rhythms (e.g., ectopy or bigeminy-like patterns) drive misclassifications, and whether additional post-processing is required. Based on these findings, we will refine the model and/or introduce lightweight decision logic such as temporal smoothing (e.g., requiring K consecutive AF-positive windows under sufficient SQI) to reduce sporadic errors. In parallel, we will finalize the deployment-oriented model simplification plan, selecting the final input representation and architecture based on the combined criteria of test performance, robustness under noise, and on-device feasibility.

In the fourth phase (April), we will complete deployment-oriented optimization and finalize the evaluation suite. This includes applying practical compression steps such as architecture simplification and quantization where appropriate, and validating that compressed models maintain consistent outputs relative to the offline reference (within an acceptable numerical tolerance). We will benchmark deployment-relevant metrics—end-to-end inference latency per update cycle (preprocessing + SQI + inference + post-processing), memory footprint, sustainable update rate, and estimated energy cost—using the chosen runtime environment. If the smartwatch execution environment supports direct inference, we will prototype on-watch inference to measure real latency and power behavior; otherwise, we will implement a watch-to-edge real-time topology (e.g., BLE streaming of windows to a paired edge runtime) while keeping the classifier wearable-class, ensuring that experimental results remain representative of eventual on-watch deployment.

In the fifth phase (May), we will consolidate final results and focus primarily on thesis writing, figure preparation, and presentation-ready documentation. We will produce final plots and tables (e.g., ROC curves, ablation comparisons of SQI gating, representative PPG segments with model outputs, and performance stratified by motion/noise), and complete the methodology and results chapters with a clear, reproducible description of the pipeline and experimental protocol. We will also reserve limited time for a small number of targeted "stretch" experiments if feasible (e.g., testing generalization on a small external dataset or evaluating an upper-bound scenario for comparison), but only after the core pipeline and reporting are complete.

In the final phase (early June, leading up to Friday 12 June 2025, 13:00), we will use buffer time for any last-minute verification runs, proofreading, formatting, and final polishing of the report, as well as preparation for the final project presentation/defense. This buffer is intended to absorb unforeseen implementation issues, ensure that all reported results are reproducible from a clean run, and guarantee on-time submission with a fully coherent narrative from problem motivation through deployment-feasible evaluation.

Chapter 7

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